

Fructokinase catalyzes the phosphorylation of fructose to fructose-1-phosphate, the first step in the metabolism of fructose in cells of the small intestine and liver. Subsequent steps produce glyceraldehyde-3-phosphate and dihydroxyacetone phosphate, which can enter glycolysis. Another enzyme, hexokinase, is found in multiple cell types and can phosphorylate both glucose and fructose, forming the glycolytic intermediates glucose-6-phosphate and fructose-6-phosphate, respectively.

Experiment 1

Six groups of wild-type (WT) mice or mice in which fructokinase had been knocked out (FK KO) throughout the body were each given water, a 10% sugar solution, or a 30% sugar solution to drink for 30 weeks. The sugar solutions contained equal amounts of glucose and fructose. During the experiment, cumulative intakes of energy and fructose from the overall diet of each mouse were determined in the groups (Table 1).

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	30% sugar	3803 $\pm$ 205	1231 $\pm$ 210
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	10% sugar	2143 $\pm$ 75	212 $\pm$ 20
	30% sugar	3026 $\pm$ 130	626 $\pm$ 33

Table 1 Cumulative intakes of energy and fructose in WT and FK KO mice.

Experiment 2

To test whether metabolism of dietary fructose by small intestinal cells affected the amount of fructose to which the liver was exposed, mutant mice lacking intestinal fructokinase were created. Mice in which fructokinase genes (*FK*) were flanked by LoxP sequences ("floxed") were crossed with a second set of mice in which Cre recombinase was expressed only in small intestinal cells. Offspring with floxed fructokinase alleles (*FK^{fl/fl}*) were studied as controls expressing small intestinal fructokinase and then compared with offspring lacking small intestinal fructokinase due to the action of Cre recombinase (*FK^{fl/fl} × Cre SI*). All mice were made to fast prior to administration of a standard experimental solution containing glucose and fructose. Portal vein fructose concentrations before and after ingestion of equal volumes of the solution were then compared between the two groups (Figure 1).

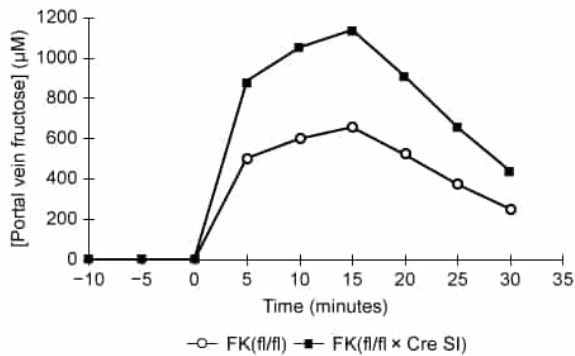


Figure 1 Portal vein fructose levels before and after ingestion of a standard glucose/fructose solution.

Following ingestion of a large amount of fructose, ATP consumption occurs more rapidly than ATP production in some cells, and the cytosolic ATP concentration of the small intestine and liver fall and then recover whereas those in skeletal muscle are relatively unchanged. Further analysis shows all three types of cells express fructose transporters. What mechanism is most likely responsible for these findings?

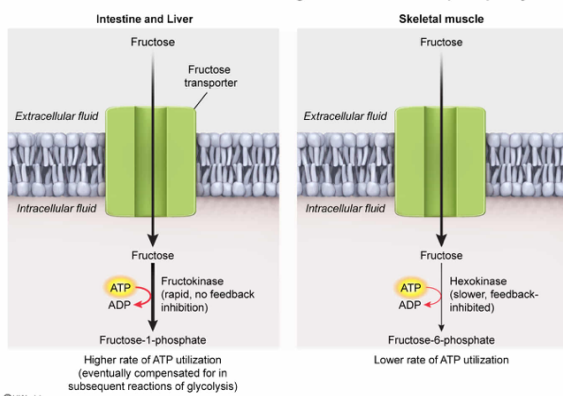
- ☐ A. Feedback inhibition of fructose phosphorylation to fructose-6-phosphate by fructokinase in small intestine and liver cells but not muscle cells [46%]
- ☐ B. Stimulation of ATP release during gluconeogenic reversal of the hexokinase reaction in muscle cells but not small intestine or liver cells [9%]
- ☒ C. Feedback inhibition of sugar phosphorylation by hexokinase in muscle cells but not small intestine or liver cells [22%]
- ☐ D. Stimulation of ATP release during gluconeogenic reversal of the fructokinase reaction in small intestine and liver cells but not muscle cells [21%]

Correct

22% answered correctly

Explanation:

Feedback inhibition of hexokinase regulates ATP use for phosphorylation



After being transported into cells, glucose and fructose are phosphorylated by **kinases** (eg, hexokinase, fructokinase). This phosphorylation "traps" (ie, prevents diffusion back

微信Mai_ange

into the extracellular fluid) the sugars inside the cells and helps maintain a favorable gradient for transport of sugars into the cell by reducing the intracellular concentration of unphosphorylated sugars.

Many metabolic **pathways are regulated** by allosteric effectors, and an effector that activates one metabolic pathway often inhibits the opposing pathway (eg, glucose-6-phosphate in glycolysis and gluconeogenesis).

In most tissues (including muscle), **allosteric feedback inhibition** of hexokinase occurs when its products (ie, glucose-6-phosphate and fructose-6-phosphate) increase in concentration. This feedback inhibition **limits ATP use for phosphorylation** in the hexokinase reaction, thereby preventing the ATP depletion that might otherwise occur. In contrast, the **fructokinase** present in small intestine and liver cells **does not undergo feedback inhibition** and **can** therefore transiently **deplete cellular ATP** levels when fructose is abundant. The ATP levels eventually recover when later steps in glycolysis produce more ATP.

(Choice A) According to the passage, fructokinase phosphorylates fructose to fructose-1-phosphate (*not* fructose-6-phosphate).

(Choice B) Although sharing some near-equilibrium reactions with glycolysis, gluconeogenesis is not a simple reversal of the reactions of glycolysis. The reaction catalyzed by hexokinase during glycolysis is not part of gluconeogenesis and is *irreversible*.

(Choice D) Phosphorylation of fructose by fructokinase generates fructose-1-phosphate in an *irreversible* reaction. Also, although the fructose-1-phosphate formed via the fructokinase reaction can be further metabolized to dihydroxyacetone phosphate and glyceraldehyde-3-phosphate, fructokinase is not a gluconeogenic enzyme.

Educational objective:

Metabolic pathways are commonly regulated by allosteric effectors, sometimes in a tissue-specific manner.

Time spent:0 sec

Subject:
Biochemistry

QID:403191

System:
Metabolic Reactions

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Table 1 Cumulative intakes of energy and fructose in WT and FK KO mice.

Experiment 2

To test whether metabolism of dietary fructose by small intestinal cells affected the amount of fructose to which the liver was exposed, mutant mice lacking intestinal fructokinase were created. Mice in which fructokinase genes (*FK*) were flanked by LoxP sequences ("floxed") were crossed with a second set of mice in which Cre recombinase was expressed only in small intestinal cells. Offspring with floxed fructokinase alleles (*FK^{fl/fl}*) were studied as controls expressing small intestinal fructokinase and then compared with offspring lacking small intestinal fructokinase due to the action of Cre recombinase (*FK^{fl/fl} × Cre SI*). All mice were made to fast prior to administration of a standard experimental solution containing glucose and fructose. Portal vein fructose concentrations before and after ingestion of equal volumes of the solution were then compared between the two groups (Figure 1).

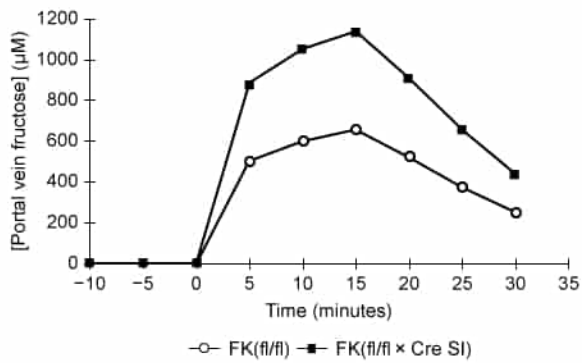


Figure 1 Portal vein fructose levels before and after ingestion of a standard glucose/fructose solution.

Which of the following enzymes involved in the small intestinal metabolism of fructose is part of the glycolytic pathway?

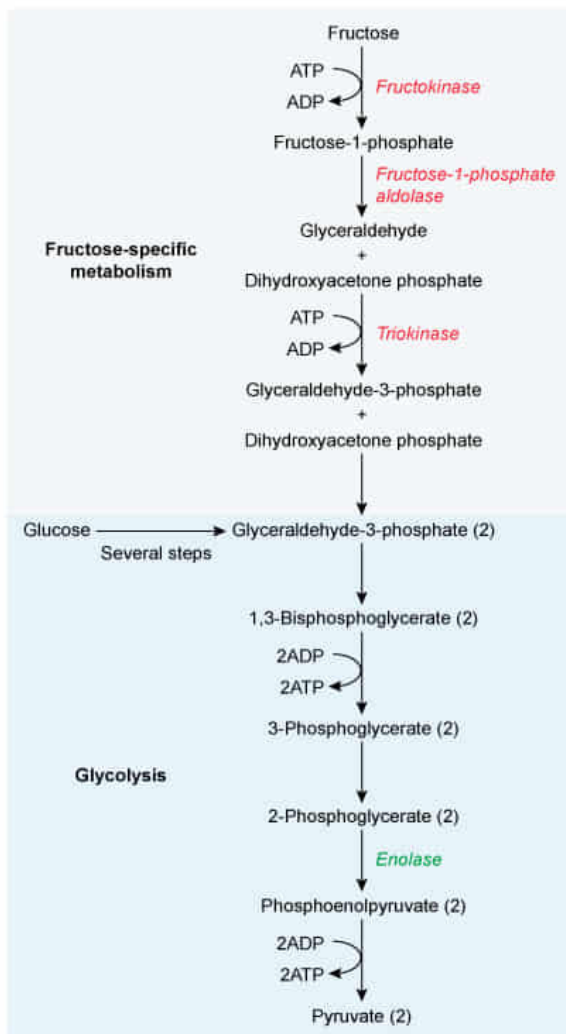
- ☐ A. Triokinase [1%]
- ☐ B. Fructose-1-phosphate aldolase [11%]
- ☐ C. Fructokinase [25%]
- ✓ ☒ D. Enolase [61%]

Correct

61% answered correctly

Explanation:

Glycolytic enzymes in fructose metabolism



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Glycolysis is the series of cytosolic reactions by which a glucose molecule is metabolized to two pyruvate molecules, generating ATP and the **reduced electron carrier** NADH in the process. The first five enzymatic reactions, in which glyceraldehyde-3-phosphate is formed and two ATP molecules are used to phosphorylate glucose and fructose-6-phosphate, constitute the "energy investment phase." The second five enzymatic reactions, during which glyceraldehyde-3-phosphate is converted to pyruvate and four ATP molecules are formed by **substrate-level phosphorylation**, are considered the "energy payoff phase."

Glucose is the parent molecule of the glycolytic pathway, but other carbohydrates (eg, polysaccharides, disaccharides, other sugars) can undergo reactions that allow them to be metabolized in glycolysis. Fructose can enter glycolysis after being phosphorylated to the glycolytic intermediate fructose-6-phosphate by hexokinase in muscle cells. Alternatively, in small intestinal and liver cells, fructose can be phosphorylated to fructose-1-phosphate by fructokinase and undergo the subsequent fructose-1-phosphate aldolase and triokinase reactions to enter glycolysis as dihydroxyacetone phosphate and glyceraldehyde-3-phosphate.

The question asks for the glycolytic enzyme that participates in fructose metabolism in small intestinal cells. Of the choices given, only **enolase**, which **catalyzes** the **formation of phosphoenolpyruvate from 2-phosphoglycerate**, is an **enzyme of glycolysis**.

(Choices A, B, and C) Triokinase, fructose-1-phosphate aldolase, and fructokinase collectively convert intracellular fructose to the glycolytic intermediates dihydroxyacetone phosphate and glyceraldehyde-3-phosphate. These reactions are necessary for fructose to enter the glycolytic pathway but they are not part of glycolysis.

Educational objective:

Glycolysis is the series of cytosolic reactions in which a glucose molecule is converted to two pyruvate molecules, resulting in the net production of two ATP molecules and two NADH molecules.

Time spent:0 sec

Subject:
Biochemistry

QID:403192

System:
Metabolic Reactions

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Table 1 Cumulative intakes of energy and fructose in WT and FK KO mice.

Experiment 2

To test whether metabolism of dietary fructose by small intestinal cells affected the amount of fructose to which the liver was exposed, mutant mice lacking intestinal fructokinase were created. Mice in which fructokinase genes (*FK*) were flanked by LoxP sequences ("floxed") were crossed with a second set of mice in which Cre recombinase was expressed only in small intestinal cells. Offspring with floxed fructokinase alleles (*FK^{fl/fl}*) were studied as controls expressing small intestinal fructokinase and then compared with offspring lacking small intestinal fructokinase due to the action of Cre recombinase (*FK^{fl/fl} × Cre SI*). All mice were made to fast prior to administration of a standard experimental solution containing glucose and fructose. Portal vein fructose concentrations before and after ingestion of equal volumes of the solution were then compared between the two groups (Figure 1).

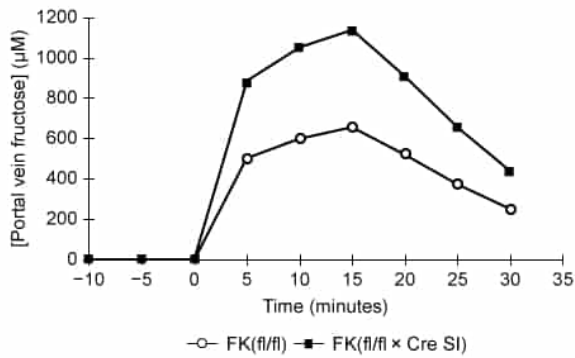


Figure 1 Portal vein fructose levels before and after ingestion of a standard glucose/fructose solution.

Which finding, when combined with data presented in Table 1, would be most likely to lead researchers to conclude that fructokinase activity mediates the abnormal accumulation of liver triglycerides in sugar-drinking mice that have similar energy and fructose intake levels?

- ☐ A. Abnormally high liver triglyceride accumulation only in WT and FK KO mice drinking the 30% sugar solution [9%]
- ☐ B. Abnormally high liver triglyceride accumulation only in WT mice consuming the 30% sugar solution [29%]
- ☒ C. Abnormally high liver triglyceride accumulation in WT mice consuming the 10% sugar solution but not in FK KO mice consuming the 30% sugar solution [46%]
- ☐ D. Abnormally high liver triglyceride accumulation in WT mice consuming the 30% sugar solution but not in FK KO mice consuming the 10% sugar solution [14%]

Correct

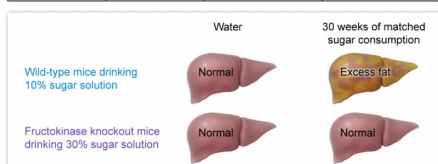
46% answered correctly

Explanation:

Fructokinase mediates the abnormal accumulation of liver triglycerides in sugar-drinking mice matched for energy and fructose intake

Fructokinase genotype	Solution	Cumulative energy intake (mean kilocalories ± standard error)	Cumulative fructose intake (mean kilocalories ± standard error)
Wild-type	Water	2251 ± 100	0
	10% sugar	2928 ± 103	640 ± 50
	30% sugar	3803 ± 205	1231 ± 210
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Energy and fructose intakes are statistically similar



Because of their roles in nutrient absorption and distribution, the small intestine and liver must handle nutrients differently than tissues receiving nutrients solely via the **systemic circulation**. Accordingly, the small intestine and liver possess special enzymes (eg, fructokinase) for phosphorylating sugars that, unlike the hexokinases found in other tissues, are not feedback-inhibited.

The passage presents cumulative energy and fructose intakes for WT and FK KO mice consuming water or sugar solutions as part of their diets. The question asks for an additional finding that would convince researchers that fructokinase mediates abnormal liver triglyceride accumulation in mice that consume similar amounts of energy and fructose.

Energy and fructose consumption in the WT mice consuming the 10% sugar solution and the FK KO mice consuming the 30% sugar solution are not statistically different (ie, values in each group are within one standard error of each other). Therefore, **abnormal triglyceride accumulation** in the group with **active fructokinase** (ie, WT mice) would provide evidence that **fructokinase activity mediates** excess **triglyceride accumulation** in sugar-drinking **mice** that consume **similar amounts of energy and fructose**.

(Choices A and D) The two groups specified in each of these choices consumed significantly different amounts of energy and fructose over the 30-week intervention period, so they would not have been compared.

(Choice B) The WT mice drinking the 30% sugar solution consumed significantly more energy and fructose than all the other groups, and therefore cannot be compared to any other group. A comparison must be made to an FK KO group to draw conclusions about effects of fructokinase activity.

Educational objective:

Different tissues can exhibit different metabolic responses that are consistent with their normal roles in whole-body metabolism.

Time spent:0 sec

Subject:
Biochemistry

QID:403193

System:
Metabolic Reactions

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Table 1 Cumulative intakes of energy and fructose in WT and FK KO mice.

Experiment 2

To test whether metabolism of dietary fructose by small intestinal cells affected the amount of fructose to which the liver was exposed, mutant mice lacking intestinal fructokinase were created. Mice in which fructokinase genes (*FK*) were flanked by LoxP sequences ("floxed") were crossed with a second set of mice in which Cre recombinase was expressed only in small intestinal cells. Offspring with floxed fructokinase alleles (*FK^{fl/fl}*) were studied as controls expressing small intestinal fructokinase and then compared with offspring lacking small intestinal fructokinase due to the action of Cre recombinase (*FK^{fl/fl} × Cre SI*). All mice were made to fast prior to administration of a standard experimental solution containing glucose and fructose. Portal vein fructose concentrations before and after ingestion of equal volumes of the solution were then compared between the two groups (Figure 1).

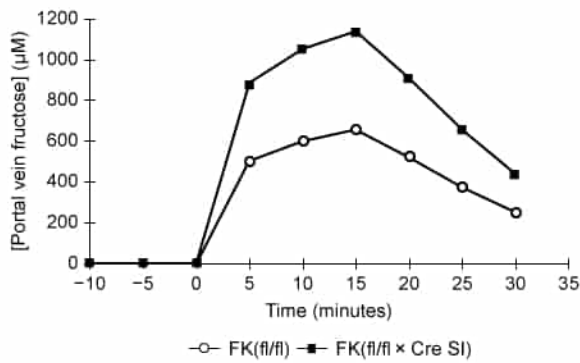


Figure 1 Portal vein fructose levels before and after ingestion of a standard glucose/fructose solution.

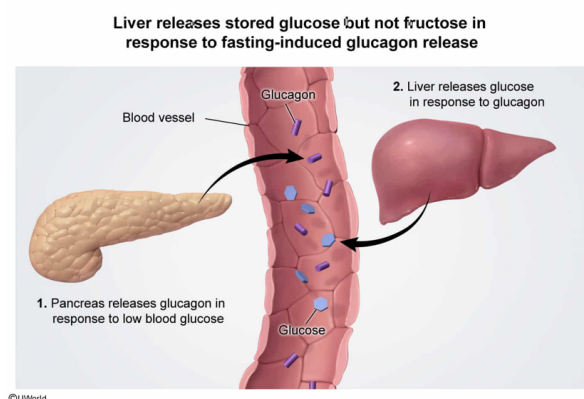
Which of the following most likely explains why fructose was NOT detected in portal vein blood prior to ingestion of the sugar solution in Experiment 2?

- ☐ A. Insulin concentrations in the fasting state were too low to promote transport of fructose into small intestine cells. [16%]
- ☐ B. Fasting cortisol levels were below the threshold required to activate the release of stored fructose. [10%]
- ☒ C. Elevated glucagon levels during fasting promoted release of stored glucose but not fructose. [71%]
- ☐ D. Increased growth hormone levels during fasting reduced glucose but not fructose uptake from the blood. [2%]

Correct

71% answered correctly

Explanation:



Nutrients in the diet are absorbed from the small intestine into the blood stream and taken from there for use by the body's tissues. During fasting periods, cells continue to take up nutrients from the blood.

In healthy individuals, blood glucose concentrations are maintained within a fairly

narrow range (around 5 mM), reflecting a general balance between glucose uptake from and entry into blood that is largely maintained by the hormones **insulin and**

glucagon. During fasting, decreasing blood glucose concentrations cause the pancreas to release glucagon. **Glucagon stimulates** the **breakdown of liver glycogen** into **glucose**, which is **released into** the **bloodstream** to balance glucose uptake from the blood throughout the body.

The question asks why fructose was not detected in the portal vein prior to fructose ingestion. During fructose ingestion, portal vein fructose concentration rises as fructose is absorbed from the small intestine. However, fructose is not stored in large quantities in the body, and unlike glucose, **fructose release into** the blood is **not stimulated** by the rise in **glucagon** during fasting. Therefore, the fasting blood fructose level will be negligible relative to that of glucose.

(Choice A) Fructose would not be present to be taken up prior to sugar ingestion, and fructose uptake by cells of the small intestine is not stimulated by insulin (insulin only stimulates glucose uptake).

(Choice B) Cortisol can stimulate glucagon secretion and thereby augment glucose (but *not* fructose) release by the liver. Therefore, cortisol would not impact fructose levels regardless of whether a certain cortisol threshold has been reached.

(Choice D) Growth hormone levels typically increase during fasting and can inhibit glucose uptake from the blood by various tissues. However, unlike glucose, there are no appreciable stores of fructose that can be drawn upon to maintain blood fructose levels after absorption.

Educational objective:

Glucagon stimulates the liver to release stored glucose into the blood.

Time spent:0 sec

Subject:
Biochemistry

QID:403194

System:
Metabolic Reactions

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Table 1 Cumulative intakes of energy and fructose in WT and FK KO mice.

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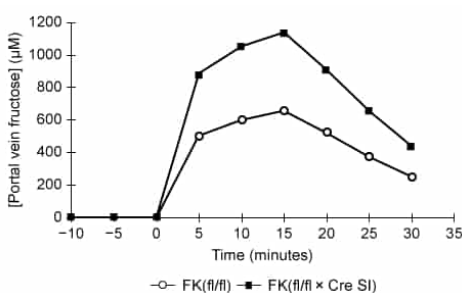


Figure 1 Portal vein fructose levels before and after ingestion of a standard glucose/fructose solution.

In Experiment 2, what is the independent variable?

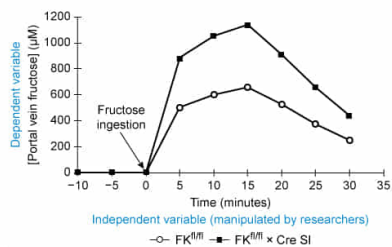
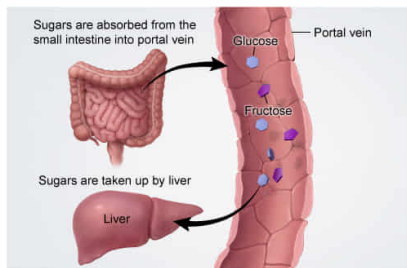
- ☐ A. Fructose concentration in the ingested solution [16%]
- ☐ B. Time at which the sugar solution was ingested [8%]
- ☐ C. Fructose concentration in the blood leaving the small intestine [18%]
- ☒ D. Time relative to ingestion of the sugar solution [55%]

Correct

55% answered correctly

Explanation:

Time relative to fructose ingestion is the independent variable



Dietary sugars enter the **systemic circulation** after being absorbed from the small intestine into the portal vein. The passage describes an experiment examining intestinal fructose absorption and metabolism in mice that lack small intestinal fructokinase ($FK^{fl/fl} \times Cre SI$ mice) and control mice that had functional intestinal fructokinase ($FK^{fl/fl}$ mice). To test whether deleting small intestinal fructokinase affected absorption of ingested fructose, portal vein fructose concentration following consumption of a standard sugar solution containing fructose was measured in the two groups of mice.

The question asks about the independent variable in the experiment. **Independent variables** are those that are **experimentally selected, varied, or controlled by a researcher**, whereas dependent variables are those proposed to vary in response to differences in the independent variable (or variables). In Experiment 2, **portal vein fructose** levels were **measured** at **various times** selected by the researcher both before and after ingestion. Therefore, **time relative to ingestion** of the sugar solution is the **independent variable** in this experiment.

(Choices A and B) These experimental factors represent *potential* independent variables, had they been experimentally varied by the researchers. However, the passage states that all mice received the same sugar solution, and there is no mention that the timing of sugar ingestion was experimentally manipulated.

Therefore, these are not independent variables in the experiment.

(Choice C) The concentration of fructose in the portal vein was unknown before the experiment and measured at specific time points determined by the researcher. Therefore, portal vein fructose concentration is the *dependent* variable in the experiment.

Educational objective:

Different tissues can exhibit different metabolic responses that are consistent with their normal roles in whole-body metabolism. Independent variables are those experimentally manipulated by researchers.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403195
System:
Metabolic Reactions

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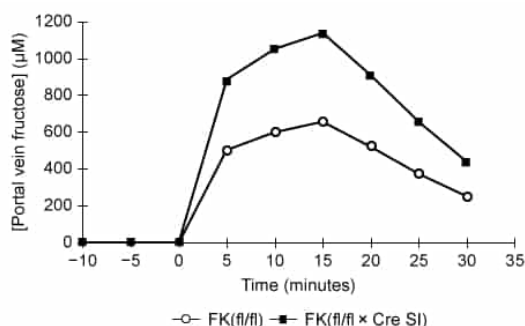


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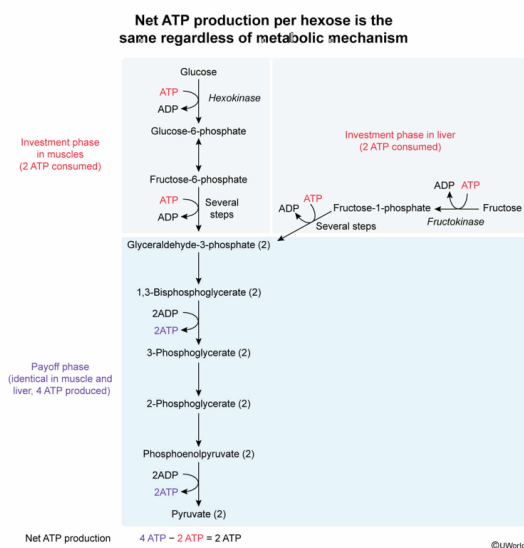
In liver cells, a molecule of fructose requires 2 ATP molecules to become phosphorylated and enter glycolysis as 2 molecules of glyceraldehyde-3-phosphate. In muscle cells, 1 molecule of fructose is initially phosphorylated by hexokinase, entering glycolysis directly as fructose-6-phosphate. Compared to net ATP production from fructose in muscle cells, net glycolytic ATP production from fructose in liver cells will:

- ☐ A. be reduced by half [27%]
☒ B. not change [47%]
☐ C. be doubled [22%]
☐ D. increase four-fold [2%]

Correct

47% answered correctly

Explanation:



The reactions of glycolysis can be divided into **two phases**:

- **Energy investment:** This phase consists of the first five reactions of glycolysis, during which glyceraldehyde-3-phosphate (GAP) is formed and **2 ATP molecules** are "invested" in phosphorylation reactions to form glycolytic intermediates necessary for the eventual production of ATP in glycolysis.
- **Energy payoff:** This phase consists of the final five reactions of glycolysis, during which GAP is converted to pyruvate, with **4 ATP molecules** (and 2 NADH that can be used for mitochondrial ATP synthesis) **formed** for each hexose sugar (ie, 6 carbons) that originally entered glycolysis.

The net ATP synthesis from 1 glucose molecule through glycolysis is therefore 2 ATP:

ATP produced during the payoff phase – ATP consumed during the investment phase = Net ATP synthesis

$$4 \text{ ATP produced} - 2 \text{ ATP consumed} = 2 \text{ ATP (net production)}$$

When fructose is phosphorylated by hexokinase and enters glycolysis directly as fructose-6-phosphate, the number of ATP molecules invested and produced in glycolysis are the same as for glucose. Fructose phosphorylation by fructokinase consumes 1 ATP, as does the downstream reaction that catalyzes the formation of the glycolytic intermediate GAP. In other

words, because 2 GAP molecules must form per hexose, 2 ATP molecules must be consumed per hexose, regardless of how those reactions happen.

The question asks for the net glycolytic ATP production from fructose in the liver cells relative to that in the muscle cells. The **number of ATP molecules invested in the formation of GAP** is the **same** (ie, 2) when fructose is phosphorylated **by fructokinase as** when fructose (or glucose) is phosphorylated **by hexokinase**, and the **energy payoff phase** in **both scenarios** is **identical to that occurring when glucose is metabolized** in glycolysis. Therefore, the **net glycolytic ATP production** will not change when fructose is metabolized via fructokinase rather than by hexokinase.

(Choices A, C, and D) These choices reflect an incorrect consideration of the number of ATP molecules invested and produced when fructose enters glycolysis following phosphorylation by either hexokinase or fructokinase.

Educational objective:

During glycolysis of 1 glucose molecule, 2 ATP molecules are invested to form intermediates that will eventually yield 4 ATP molecules, representing a net production of 2 ATP molecules in the pathway.

Time spent:0 sec

Subject:
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System:
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